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EXPERIENCE

Sprints AI

Research Engineer - R&D

Cairo, EG
Sep 2025 – Present

- Architected and implemented a multi-tenant AI Learning Companion platform with production FastAPI services, org-scoped authentication, Supabase/Postgres persistence, Qdrant retrieval, and Redis-backed state management.
- Built a precision content-ingestion pipeline for videos, PDFs, code, markdown, and curriculum maps, preserving timestamps, spatial coordinates, line ranges, and hierarchy metadata to support exact citations and navigation.
- Developed production agent workflows for tutoring, career coaching, study tools, and educational games, combining persona-based orchestration, curriculum-aware context, hybrid retrieval, and backend-switchable Qdrant/HelixDB search.
- Implemented neuroscience-inspired memory, guardrails, and resource-safety layers for personalized learning flows, including episodic/semantic/procedural memory, preference recall, and VPS-safe ingestion under constrained infrastructure.

Dark Entry

Research Engineer (Contractor)

Jun 2025 – Dec 2025

- Architected production platforms and ingestion orchestrators handling 50+ concurrent feeds, featuring custom adapters, SHA256 fingerprinting, and Redis-backed queuing.
- Developed a dual-implementation (Python/Rust) data processing engine using Rayon and Crossbeam for 3x throughput gains via tiered pattern matching.
- Built a health-tracking system with configurable cooldowns and self-healing auto-recovery logic, reducing manual intervention by 80%.
- Implemented distributed system patterns including Circuit Breaker state machines, adaptive rate limiters, and semaphore-based concurrency control for external APIs.
- Refactored monolithic services into modular FastAPI architectures, reducing database queries by 70% via bulk operations and NDJSON streaming for multi-terabyte datasets.
- Hardened infrastructure with Docker/Compose, Celery/Redis task orchestration, and structured observability using Prometheus metrics and OpenTelemetry tracing.

Freelance Consulting & Contracts

ML Engineer (part-time) - (selected projects alongside full-time roles)

Remote
Sep 2023 – Present

- Architected a multi-agent research framework using MCP servers and sequential pipelines to orchestrate autonomous information gathering and synthesis workflows.
- Designed and built an autonomous NER pipeline for unstructured data using spaCy and transformer models, featuring responsible AI guardrails, continuous fine-tuning via active learning, and automated ingestion.
- Developed and benchmarked oblique decision tree (ODT) controllers for pharmaceutical process control, extracting explicit control laws and comparing learned decision regions against MPC baselines.
- Built a pharmaceutical process optimization framework using GP, DKL, BART, and CVAE models for experiment selection, design-space analysis, and inverse recipe generation.
- Implemented a multi-process async benchmark system (for Stakpak) to evaluate and optimize LLMs, boosting experimentation velocity by 3x and enabling seamless provider switching across OpenAI, Anthropic, GDM and open-source models.

Proteinea

Deep Learning Intern

Cairo, EG
Aug 2022 – Nov 2022

- Developed initial T5-based models for antigen-antibody interactions, laying foundation for graduation thesis that later surpassed SoTA baselines.
- Automated evaluation pipelines and created TensorBoard visualizations, reducing analysis time by 30%.

PROJECTS

Antigen-Antibody Translation/Generation – PyTorch, HuggingFace, T5, Pandas, AWS

Graduation Research Thesis

- Developed a custom T5 Transformer that achieved 10% higher performance than 2022 SoTA baselines on the SABDAB dataset (building on insights from Proteinea internship) using AWS NVIDIA A10G GPUs.
- Enabled real-time monitoring with TensorBoard dashboards and facilitated team collaboration using Termius Vault and Tmux.

Multi-Agent Research System – Google ADK, Pydantic, Asyncio, FastAPI, Gradio

- Migrated a research workflow to Google’s Agent Development Kit (ADK), creating a SequentialAgent pipeline for data collection, analysis, and report synthesis.
- Integrated external APIs (BraveSearch, ArXiv, Wikipedia) and SQLite persistence to enhance system modularity and data gathering capabilities.
- Drafted architecture diagrams and communication patterns to streamline development and ensure scalability.

LiveMem / THEN – PyTorch, nanochat, NumPy, Attention Hooks

Independent Research

- Architected a stateful “live memory” extension for nanochat (THEN architecture) that decouples weight training from post-training episodic memory ingestion and query-time recall.
- Implemented a disk-tiered memory manager using `numpy.memmap` and chunked top-k retrieval to stream long-horizon traces from disk with bounded device-memory usage.
- Built ingestion and query pipelines for frozen-model memory population and stateful recall, including persistence validation and synthetic long-horizon episode generation for benchmarking.

SKILLS

Programming Languages: Python, Rust, C++, Triton.

Frameworks & Libraries: PyTorch, TensorFlow, FastAPI, SQLAlchemy, Celery, Redis, Scrapy, Selenium, Playwright, Pydantic, Pandas, NumPy, Hugging Face, Scikit-Learn, OpenTelemetry, Prometheus, structlog, bun, NextJS

Domain skills: Distributed Systems, System Architecture, ML/DL, NLP (NER, LLMs, RAG), Computer Vision, Generative AI, Statistical Inference, Data Analysis, Vector Databases (Qdrant, Milvus, Helix)

Tools & Platforms: Git, Docker, AWS, GCP, Azure, Ollama, Linux, WSL, SQLite, REST APIs, Kaggle, PostgreSQL

EDUCATION

University of Science and Technology – Zewail City

Bachelor of Science, Communication and Information Engineering

Giza, EG
Jun 2024